BLOCKCHAIN TECHNILOGY(UNIT-1)

A blockchain is a type of *distributed* database or ledger—one of today's top tech trends—which means the power to update a blockchain is distributed between the nodes, or participants, of a public or private computer network. This is known as distributed ledger technology.

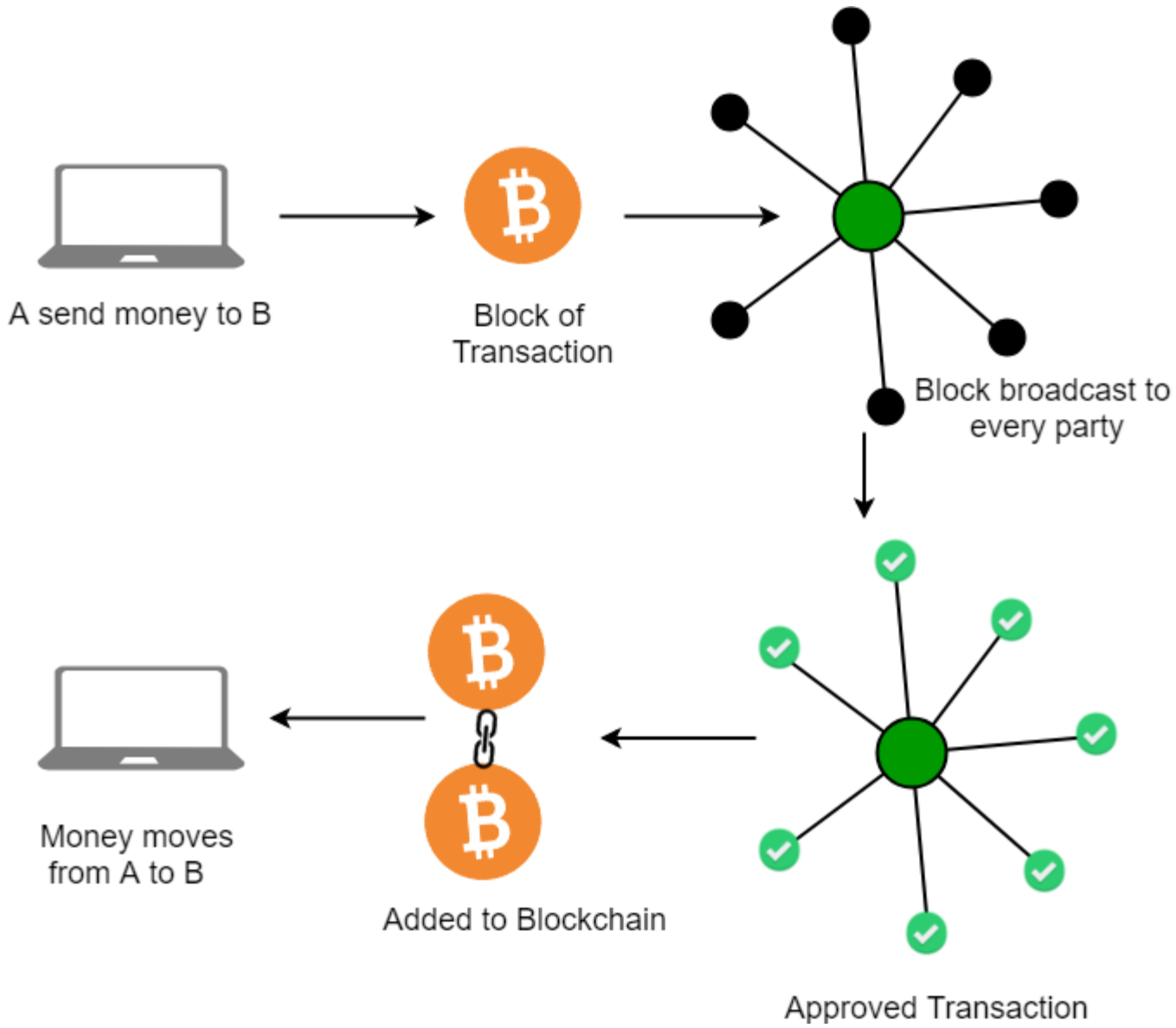
Blockchain allows for the permanent, immutable, and transparent recording of data and transactions. This, in turn, makes it possible to exchange anything that has value, whether that is a physical item or something less tangible.

A blockchain has three central attributes. First, a blockchain database must be cryptographically secure. That means in order to access or add data on the database, you need two cryptographic keys: a public key, which is basically the address in the database, and the private key, which is a personal key that must be authenticated by the network.

Next, a blockchain is a *digital* log or database of transactions, meaning it happens fully online.

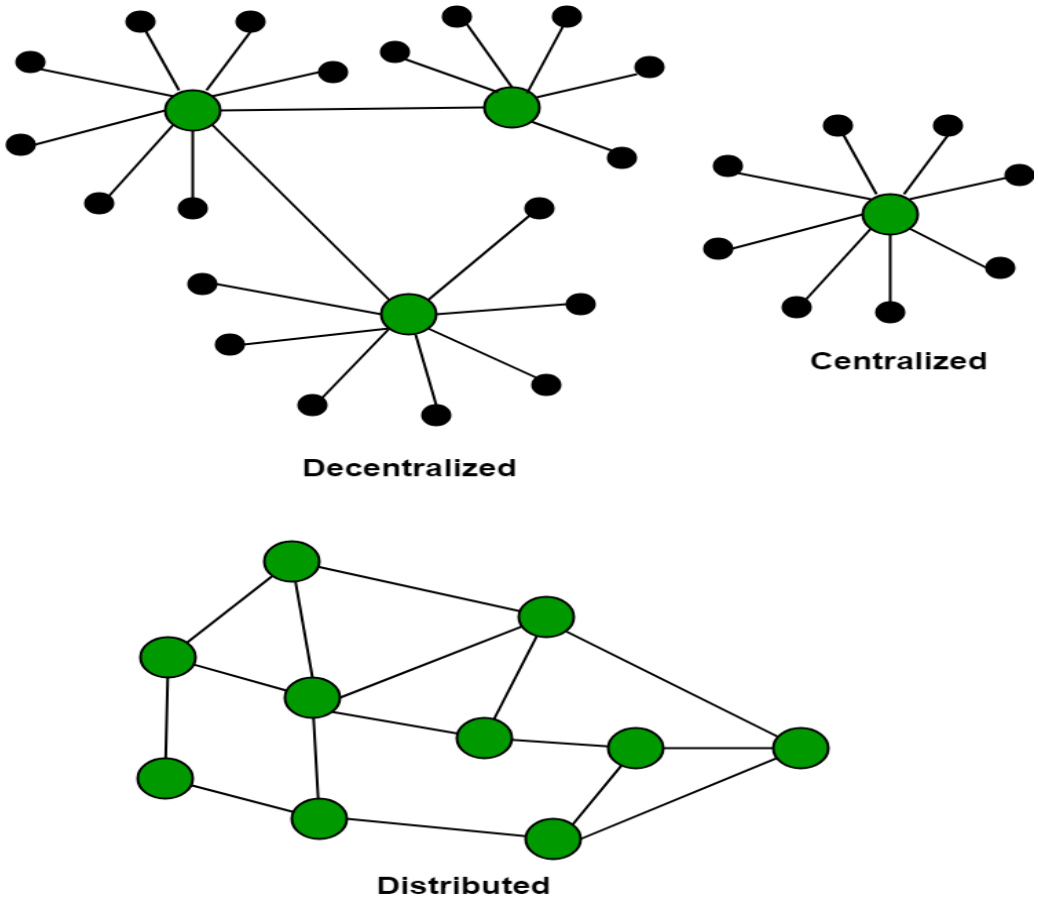
How does Blockchain Technology work?

One of the famous use of Blockchain is Bitcoin. Bitcoin is a cryptocurrency and is used to exchange digital assets online. Bitcoin uses cryptographic proof instead of third-party trust for two parties to execute transactions over the internet. Each transaction protects through digital signature.



Distributed Database: There is no Central Server or System which keeps the data of the Blockchain. The data is distributed over Millions of Computers around the world which are connected to the Blockchain. This system allows

Notarization of Data as it is present on every Node and is publicly



verifiable.

Benefits of Blockchain Technology:

- **Time-saving:** No central Authority verification needed for settlements making the process faster and cheaper.
- **Cost-saving:** A Blockchain network reduces expenses in several ways. No need for third-party verification. Participants can share assets directly. Intermediaries are reduced. Transaction efforts are minimized as every participant has a copy of shared ledger.
- **Tighter security:** No one can temper with Blockchain Data as it shared among millions of Participant. The system is safe against cybercrimes and Fraud.
- **Collaboration:** It permits every party to interact directly with one another while not requiring third party negotiate.
- **Reliability:** Blockchain certifies and verifies identities of every interested party. This removes double record, reducing rates and accelerate transactions.

Application of Blockchain

- Leading Investment Banking Companies like Credit Suisse, JP Morgan Chase, Goldman Sachs and Citigroup have invested in Blockchain and are experimenting to improve the banking experience and secure it.
- Following the Banking Sector, the Accountants are following the same path. Accountancy involves extensive data, including financial statements spreadsheets containing lots of personal and institutional data. Therefore, accounting can be layered with blockchain to easily track confidential and sensitive data and reduce human error and fraud. Industry Experts from Deloitte, PwC, KPMG and EY are proficiently working and using blockchain-based software.
- Booking a Flight requires sensitive data ranging from the passenger's name, credit card numbers, immigration details, identification, destinations, and sometimes even accommodation and travel information. So the sensitive data can be secured using blockchain technology. Russian Airlines are working towards the same.
- Various industries, including hotel services, pay a significant amount ranging from 18-22% of their revenue to third-party agencies. Using blockchain, the involvement of the middleman is cut short and allows interacting directly with the consumer ensuring benefits to both parties. Winding Tree works extensively with Lufthansa, AirFrance, AirCanada, and Etihad Airways to cut short third-party operators charging high fees.
- Barclaysuses Blockchain to streamline the Know Your Customer (KYC) and Fund Transfer processes while filling patents against these features.
- Visauses Blockchain to deal with business to business payment services.
- Unileveruses Blockchain to track all their transactions in the supply chain and maintain the product's quality at every stage of the process.
- Walmart has been using Blockchain Technology for quite some time to keep track of their food items coming right from farmers to the customer. They let the customer check the product's history right from its origin.
- DHL and Accenture working together to track the origin of medicine until it reaches the consumer.
- Pfizer,an industry leader, has developed a blockchain system to keep track of and manage the inventory of medicines.
- The government of Dubai looking forward to making Dubai the first-ever city to rely on entirely and work using blockchain, even in their government office.
- Along with the above organisations, leading tech companies like Google, Microsoft, Amazon, IBM, Facebook, TCS, Oracle, Samsung, NVIDIA, Accenture, PayPal, are working on Blockchain extensively.

HISTORY OF BLOCKCHAIN TECHNOLOGY

1. In 1991, Stuart Haber and W. Scott Stornetta ideated the concept of a cryptographically secured chain of blocks. In 1992 they used Merkle trees to create a chain ofeach connected to the one before it. Newest record in this chain would contain the history of the entire chain. Note: is named after Ralph Merkle who patented them in 19792. In 2004, computer scientist and cryptographic activist Hal Finney introduced a system called Reusable Proof Of Work(RPoW) as a prototype for digital cash. It was a significant early step inthe history of crypto-currencies. 3. Further in 2008, Satoshi Nakamato conceptualized the theory of distributed block chain where The modified trees would contain a secure history of data exchanges, utilize a peer-topeer network for time stamping and verifying each exchange, and could be manage dautonomously without a central authority. The design serves as the public ledger for all transactions in the crypto-currency space.

Distributed Ledger Technology (DLT) is centered around an encoded and distributed database where records regarding transactions are stored. A distributed ledger is a database that is spread across various computers, nodes, institutions, or countries accessible by multiple people around the globe.
Features:

1. **Decentralized:** It is a decentralized technology and every node will maintain the ledger, and if any data changes happen, the ledger will get updated. The process of updating takes place independently at each node. Even small updates or changes made to the ledger are reflected and the history of that change is sent to all participants in a matter of seconds.
2. **Immutable:** Distributed ledger uses cryptography to create a secure database in which data once stored cannot be altered or changed.
3. **Append only:** Distributed ledgers are append-only in comparison to the traditional database where data can be altered.
4. **Distributed:** In this technology, there is no central server or authority managing the database, which makes the technology transparent. To counter the weaknesses of having one ledger to rule all, So that there is no one authoritative copy and have specific rules around changing them. This would make the system much more transparent and will make it a more decentralized authority. In this process, every node or contributor of the ledger will try to verify the transactions with the various consensus algorithms or voting. the voting or participation of all the nodes depends on the rules of that ledger. In the case of bitcoin, the Proof of Work consensus mechanism is used for the participation of each node.
5. **Shared:** The distributed ledger is not associated with any single entity. It is shared among the nodes on the network where some nodes have a full copy of the ledger

while some nodes have only the necessary information that is required to make them functional and efficient.

Types of Blockchain

Public blockchain

How it works. The first type of blockchain technology is public blockchain. This is where cryptocurrency like Bitcoin originated and helped to popularize distributed ledger technology (DLT). It removes the problems that come with centralization, including less security and transparency. DLT doesn't store information in any one place, instead distributing it across a peer-to-peer network. Its decentralized nature requires some method for verifying the authenticity of data. That method is a consensus algorithm whereby participants in the blockchain reach agreement on the current state of the ledger. Proof of work (PoW) and proof of stake (PoS) are two common consensus methods.

Public blockchain is non-restrictive and permissionless, and anyone with internet access can sign on to a blockchain platform to become an authorized node. This user can access current and past records and conduct mining activities, the complex computations used to verify transactions and add them to the ledger. No valid record or transaction can be changed on the network, and anyone can verify the transactions, find bugs or propose changes because the source code is usually open source.

Advantages. One of the advantages of public blockchains is that they are completely independent of organizations, so if the organization that started it ceases to exist the public blockchain will still be able to run, as long as there are computers still connected to it. "Some blockchains incentivize users to commit computer power to securing the network by providing a reward," noted James Godefroy, a senior manager at Rouse, an intellectual property services provider.

Another advantage of public blockchains is the network's transparency. As long as the users follow security protocols and methods fastidiously, public blockchains are mostly secure.

2. Private blockchain

How it works. A blockchain network that works in a restrictive environment like a closed network, or that is under the control of a single entity, is a private blockchain. While it operates like a public blockchain network in the sense that it uses peer-to-peer connections and decentralization, this type of blockchain is on a much smaller scale. Instead of just anyone being able to join and provide computing power, private blockchains typically are operated on a small network inside a company or organization. They're also known as permissioned blockchains or enterprise blockchains.

Advantages. The controlling organization sets permission levels, security, authorizations and accessibility. For example, an organization setting up a private blockchain network can determine which nodes can view, add or change data. It can also prevent third parties from accessing certain information.

"You can think of private blockchains as being the intranet, while the public blockchains are more like the internet," Godefroy said.

Because they're limited in size, private blockchains can be very fast and can process transactions much more quickly than public blockchains.

Disadvantages. The disadvantages of private blockchains include the controversial claim that they aren't true blockchains, since the core philosophy of blockchain is decentralization. It's also more difficult to fully achieve trust in the information, since centralized nodes determine what is valid. The small number of nodes can also mean less security. If a few nodes go rogue, the consensus method can be compromised.

Additionally, the source code from private blockchains is often proprietary and closed. Users can't independently audit or confirm it, which can lead to less security. There is no anonymity on a private blockchain, either.

Hybrid blockchain

How it works. Sometimes, organizations will want the best of both worlds, and they'll use hybrid blockchain, a type of blockchain technology that combines elements of both private and public blockchain. It lets organizations set up a private, permission-based system alongside a public permissionless system, allowing them to control who can access specific data stored in the blockchain, and what data will be opened up publicly.

Typically, transactions and records in a hybrid blockchain are not made public but can be verified when needed, such as by allowing access through a smart contract. Confidential information is kept inside the network but is still verifiable. Even though a private entity may own the hybrid blockchain, it cannot alter transactions.

When a user joins a hybrid blockchain, they have full access to the network. The user's identity is protected from other users, unless they engage in a transaction. Then, their identity is revealed to the other party.

Advantages. One of the big advantages of hybrid blockchain is that, because it works within a closed ecosystem, outside hackers can't mount a 51% attack on the network. It also protects privacy but allows for communication with third parties. Transactions are cheap and fast, and it offers better scalability than a public blockchain network.

Disadvantages. This type of blockchain isn't completely transparent because information can be shielded. Upgrading can also be a challenge, and there is no incentive for users to participate or contribute to the network.

Consortium blockchain

How it works. The fourth type of blockchain, consortium blockchain, also known as a federated blockchain, is similar to a hybrid blockchain in that it has private and public blockchain features. But it's different in that multiple organizational members collaborate on a decentralized network. Essentially, a consortium blockchain is a private blockchain with limited access to a particular group, eliminating the risks that come with just one entity controlling the network on a private blockchain.

In a consortium blockchain, the consensus procedures are controlled by preset nodes. It has a validator node that initiates, receives and validates transactions. Member nodes can receive or initiate transactions.

Advantages. A consortium blockchain tends to be more secure, scalable and efficient than a public blockchain network. Like private and hybrid blockchain, it also offers access controls.

Disadvantages. Consortium blockchain is less transparent than public blockchain. It can still be compromised if a member node is breached, the blockchain's own regulations can impair the network's functionality.

What is decentralization?

Decentralization in blockchain refers to the transfer of authority and decision-making from a centralized entity (person, organization, or group thereof) to a distributed network.

Decentralized networks are designed to reduce the amount of trust that members must have in one another and to prevent members from abusing power or authority in ways that undermine the network's functionality.

Types of decentralization in blockchain

Before we can comprehend the many sorts of decentralization in blockchain, we must first comprehend the various levels of decentralization in general. Decentralization levels are covered further down.

Fully Centralized − A system in which the whole system is controlled and managed by a single central authority. Take, for instance, the banking system.

Semi-decentralized − A system in which the entire system is controlled and managed by numerous intermediaries.

Fully Decentralized − A system in which no middlemen are used to govern or administer it. Take Bitcoin, for example.

Architectural Decentralization − This sort of decentralization is concerned with the number of physical computers in the system. And how many PCs can it withstand malfunctioning at the same time?

As a result, architectural decentralized blockchains are ones in which the same blockchain is run by various systems. The blockchain will now be unaffected even if one

machine crashes. Bitcoin is a well-known example of decentralized blockchain architecture.

Political Decentralization − This sort of decentralization refers to the number of people or groups in charge of or managing the computers in a system.

Bitcoin, for example, is politically decentralized since it is not owned by any organization or individual. The Bitcoin protocol is used by all nodes in the Bitcoin network.

Logical Decentralization − This sort of decentralization is concerned with the representation of the system's interface and data structures. Bitcoin, for example, is not theoretically decentralized since it has a single agreed-upon state and operates as a single system.

Decentralization's Advantages

Following are the advantages of decentralization −

Provides a trustless environment − No one has to know or trust anybody other in a decentralized blockchain network. In the form of a distributed ledger, each member of the network owns a copy of the exact same data. If a member's ledger is tampered with or corrupted in any manner, the majority of the network's members will reject it.

Increases the accuracy of data reconciliation − Companies frequently share information with their partners. This data is then changed and kept in each party's data silos, only to be resurfaced when it's time to transfer it downstream. Each time data is converted, the possibility of data loss or inaccurate data entering the workstream increases.

Points of vulnerability are reduced − Decentralization can help to mitigate sources of vulnerability in systems when single actors are overly reliant. Systemic failures might result from these flaws, such as the inability to provide promised services or inefficient service owing to resource exhaustion, recurrent outages, bottlenecks, a lack of appropriate incentives for effective service, or corruption.

Distributes resources more efficiently − Decentralization may also aid in resource distribution optimization, ensuring that promised services are delivered with improved performance and consistency, as well as a lower risk of catastrophic failure.

Transparent − Because decentralized blockchains are available to the public, they are transparent. The blockchain is open to everyone with an internet connection. Each participating node keeps a single copy of the data.

Full Control − With decentralization, the blockchain's members or users have complete authority over the activities. Because there is no central authority, all of the blockchain's data, control, and power are in the hands of its users.

Immutable − The data contained in a blockchain in a decentralized blockchain is nearly hard to alter. Because each alteration must be confirmed by each node in the blockchain network, this is the case.

Secure − Decentralized blockchains are far more secure than centralized blockchains because they employ encryption to protect data. Furthermore, the data in the current block requires data from the preceding block to be cryptographically confirmed.

Disadvantages of decentralization in Blockchain

Following are the disadvantages of decentralization −

Cost − In an organization, decentralization might be costlier than centralization. Because it necessitates the development of communication-automation systems and technologies.

Conflict − decentralization should only be employed when the consumers' needs are met. Because disputes might arise if users do not properly preserve decentralization.

Volatility − Cryptocurrencies built on a decentralized blockchain are extremely volatile. This is due to the fact that cryptocurrencies, or possibly the entire technology, are relatively new to the market. As a result, a large number of individuals are investing in them.

Crime − This is due to the fact that everything is done on the network anonymously, which might lead to exploitation or misuse.

Platforms for decentralization

Today, there are many platforms available for decentralization. In fact, the fundamental feature of blockchain networks is to provide decentralization. Therefore, any blockchain network, such as Bitcoin, Ethereum, Hyperledger Fabric, or Quorum, can be used to provide a decentralization service. Many organizations around the world have introduced platforms that promise to make distributed application development easy, accessible, and secure. Some of these platforms are described as follows.

Ethereum

Ethereum tops the list as being the first blockchain to introduce a Turing-complete language and the concept of a virtual machine. This is in stark contrast to the limited scripting language in Bitcoin and many other cryptocurrencies. With the availability of its Turing-complete language, Solidity, endless possibilities have opened for the

development of decentralized applications. This blockchain was first proposed in 2013 by Vitalik Buterin, and it provides a public blockchain to develop smart contracts and decentralized applications. Currency tokens on Ethereum are called **ethers**.

Introduction to smart contracts

Smart contracts are the fundamental building blocks of Ethereum applications. They are computer programs stored on the blockchain that allow converting traditional contracts into digital parallels. Smart contracts are very logical - following an if this then that structure. This means they behave exactly as programmed and cannot be changed.

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Nick Szabo coined the term "smart contract". In 1994, he wrote an introduction to the concept(opens in a new tab)↗ and, in 1996, an exploration of what smart contracts could do(opens in a new tab)↗.

Nick Szabo envisioned a digital marketplace built on these automatic, cryptographically secure processes. A place where transactions and business functions can happen trustlessly — without intermediaries. Smart contracts on Ethereum put this vision into practice.

What are contracts?

You're probably thinking: "*I'm not a lawyer! Why would I care about contracts?*". For most people, contracts bring to mind needlessly long terms and conditions agreements or boring legal documents.

Contracts are just agreements. That is, any form of agreement can be encapsulated within the conditions of a contract. Verbal agreements or pen-and-paper contracts are acceptable for many things, but they aren't without flaws.

Trust and contracts

One of the biggest problems with a traditional contract is the need for trusted individuals to follow through with the contract's outcomes.

Here is an example:

Alice and Bob are having a bicycle race. Let's say Alice bets Bob $10 that she will win the race. Bob is confident he'll be the winner and agrees to the bet. In the end, Alice finishes the race well ahead of Bob and is the clear winner. But Bob refuses to pay out on the bet, claiming Alice must have cheated.

This silly example illustrates the problem with any non-smart agreement. Even if the conditions of the agreement get met (i.e. you are the winner of the race), you must still trust another person to fulfill the agreement (i.e. payout on the bet).

Smart contracts

Smart contracts digitize agreements by turning the terms of an agreement into computer code that automatically executes when the contract terms are met.

A digital vending machine

A simple metaphor for a smart contract is a vending machine, which works somewhat similarly to a smart contract - specific inputs guarantee predetermined outputs.

- You select a product
- The vending machine returns the amount required to purchase the product
- You insert the correct amount
- The vending machine verifies you have inserted the correct amount
- The vending machine dispenses the product of choice

The vending machine will only dispense your desired product after all requirements are met. If you don't select a product or insert enough money, the vending machine won't give out your product.

Automatic execution

One of the most significant benefits smart contracts have over regular contracts is that the outcome is automatically executed when the contract conditions are realized. There is no need to wait for a human to execute the result. In other words: smart contracts remove the need for trust.

For example, you could write a smart contract that holds funds in escrow for a child, allowing them to withdraw funds after a specific date. If they try to withdraw the funds before the specified date, the smart contract won't execute. Or, you could write a contract that automatically gives you a digital version of a car's title when you pay the dealer.

Predictable outcomes

The human factor is one of the biggest points of failure with traditional contracts. For example, two individual judges may interpret a traditional contract in different ways. Their interpretations could lead to different decisions getting made and disparate outcomes. Smart contracts remove the possibility of different interpretations. Instead, smart contracts execute precisely based on the conditions written within the contract's code. This

precision means that given the same circumstances, the smart contract will produce the same result.

Public record

Smart contracts are also useful for audits and tracking. Since Ethereum smart contracts are on a public blockchain, anyone can instantly track asset transfers and other related information. You can check to see that someone sent money to your address, for example.

Privacy protection

Smart contracts can also protect your privacy. Since Ethereum is a pseudonymous network (your transactions are tied publicly to a unique cryptographic address, not your identity), you can protect your privacy from observers.

Visible terms

Finally, like contracts, you can check what's in a smart contract before you sign it (or otherwise interact with it). Better yet, public transparency of the terms in the contract means that anyone can scrutinize it

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